Communication:	**Overall:**
Language: easy to understand, answers question, follows instructions, concise, precise	
Chemistry: proper vocab, notation, spacing, logical	vague/basic/OK/good/clearly written
Math: organized, units, sig digs, math notation, concluding statement, spacing	

SCH4U

Name: ____________________________

Monday September 29, 2025

Unit 1: Structures & Properties Test (AM)

LL: /4	Comm: /10	Total: /40

1. [2 marks] Agree or disagree with this statement & explain why:

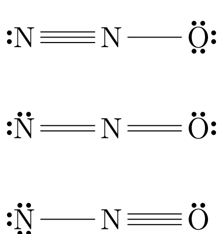
"Hydrogen only produces 4 lines in its emission spectrum"

2. [3 marks] Which of the following sets of quantum numbers (n, ℓ, m_ℓ, m_s) are possible? If it's possible, which element's last electron has this quantum number?

a.) $4, 0, 0, +\frac{1}{2}$ **b.)** $3, 2, -3, +\frac{1}{2}$ **c.)** $2, 3, 0, -\frac{1}{2}$

3. [2 marks] Suggest a possible shorthand electron configuration for neutral Rh (element 45). Is this configuration anomalous? Explain.

4. [3 marks] Using formal charge, identify which structure is **least** likely. Explain your choice.



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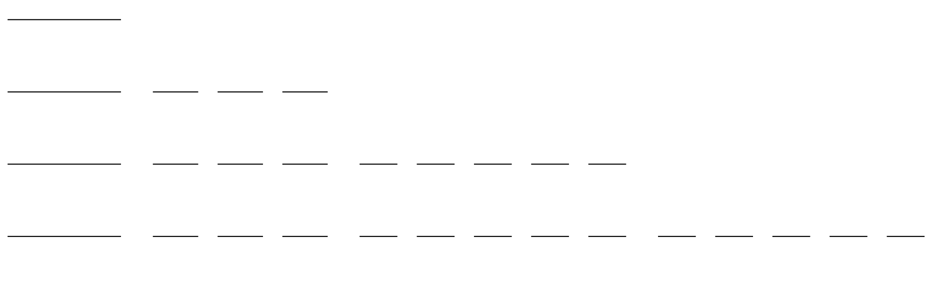
5. For the element molybdenum (element 42):

   a. [2 marks] Predict the most stable cation for molybdenum. Explain briefly.

   b. [2 marks] Draw the energy level diagram of the most stable cation from (a). Label all subshells.



6. Some atoms have anomalous electron configurations:

   a. [2 marks] List all the atoms with anomalous configurations in the 3d subshell.

   b. [2 marks] Is silicon likely to have an anomalous configuration? Explain.

7. [2 marks] Why is anomalous electron configuration not the same as hybridization? Provide an example.

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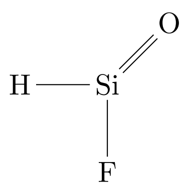
8. [4 marks] Draw the orbital overlap showing the bonding in the following structure. Use different colours for each bond (if you have them). Label each bond as being σ or π.



9. [2 marks] What is the same & what is different between the following: AB_2E & AB_2E_2? Be precise.

10. [2 marks] Why are ionic solids brittle? What is unique about the structure that allows it to be so? Provide an example of an ionic solid.

11. For the compound SF_4

   a. [1 mark] Write the VSEPR notation ____________________

   b. [1 mark] Draw the 3D ball & stick model

c. [2 marks] Is the molecule polar? Explain

d. [1 mark] Circle the intermolecular force(s) LF dd H-bond

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12. [2 marks] Element X releases red light when an e falls from $n = 5$ to $n = 3$. Element Y releases violet light when an electron falls from $n = 5$ to $n = 3$. Which element would you predict has the larger radius from nucleus to the outermost electrons? Explain briefly.

Multiple Choice:

Put your answers to the multiple choice in the table.

1.	2.	3.	4.	5.

1. Which of the following represents the ground state electron configuration of the Zn^{2+} ion?

   A. $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^{10} 4p^2$
   B. $1s^2 2s^2 2p^6 3s^2 3p^6 3d^{10}$
   C. $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 4p^6 3d^8$
   D. $1s^2 2s^2 2p^6 3s^2 3p^6 3d^8 4p^2$
   E. $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^8$

2. Which statement about bonding orbitals is FALSE?

   A. the overlap of two s-orbitals forms a σ-bond
   B. the overlap of one sp^2 orbital & one p-orbital forms a π-bond
   C. the overlap of two sp^2 orbitals forms a σ-bond
   D. the overlap of two p-orbitals form a σ-bond
   E. all are false

3. In a molecule of C_2H_4, what types of orbitals form bonds around each **carbon** atom?

   A. two s-orbitals & two p-orbitals
   B. four sp^3 orbitals
   C. four sp^2 orbitals
   D. one p-orbital & three sp^3-orbitals
   E. one p-orbital & three sp^2-orbitals

4. Which of the following molecules have polar bonds but are non-polar molecules?

I. BeH_2 **II.** BH_3 **III.** H_2O **IV.** NH_3 **V.** CCl_4

A. I & II **B.** I & V **C.** II & III **D.** II, III & IV **E.** I, II & V

5. What is the correct notation for a molecule containing a central atom attached to three other atoms with two lone pairs of electrons?

A. AB_5 **B.** AB_3E_2 **C.** AB_2E_3 **D.** AE_5 **E.** AB_5E_3